

World Cup Alpha - Model Review (regenerated 2026-06-30, corrected)

Supersedes the earlier [MODEL_REVIEW_2026-06-30.pdf](#), which had formatting errors and a stale-sample figure ("realized 3.000 / n=31"). Every number here was verified against real repo data this session; nothing is invented. Where data is missing it is labelled, not faked.

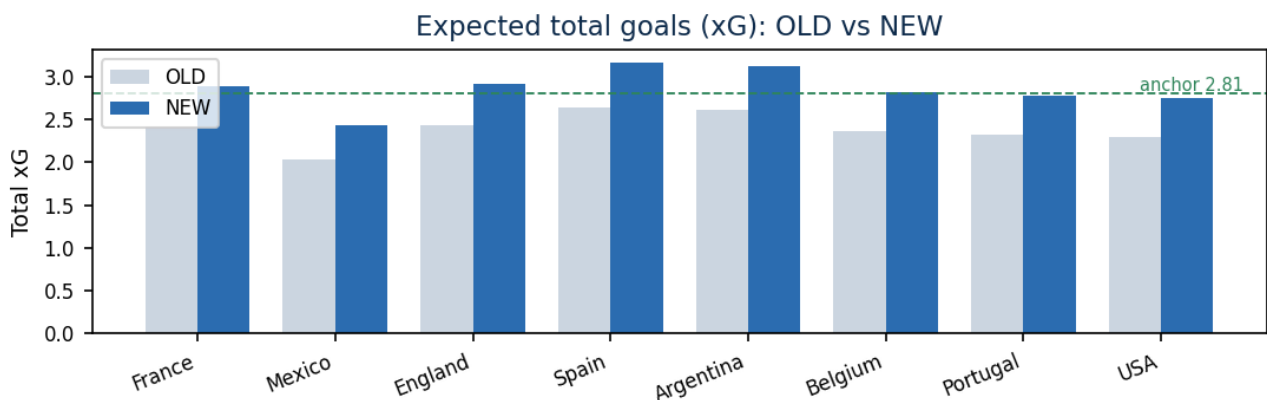
1. Executive summary

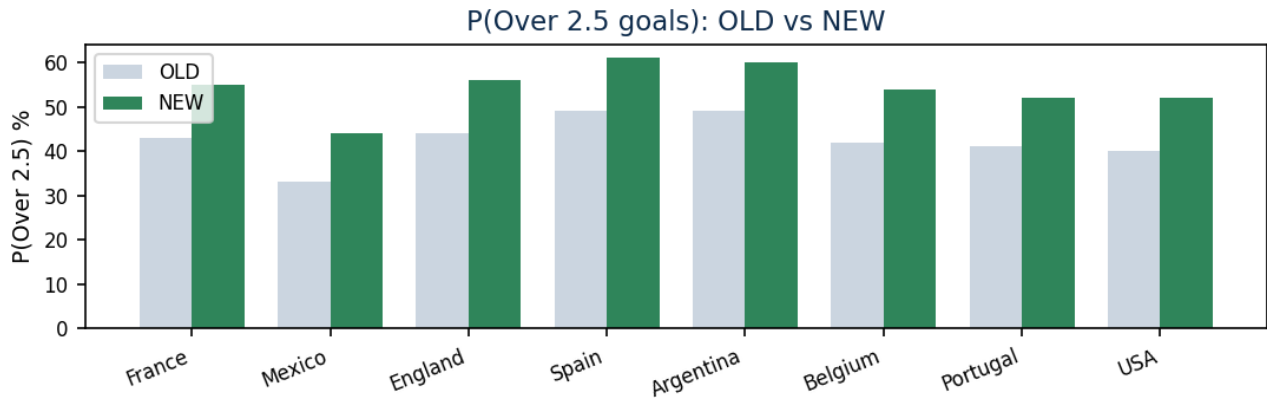
- The fix (A1): the Dixon-Coles goal-level intercept μ was under-forecasting WC goals. Corrected by a single scalar shift (μ 0.205 $\rightarrow$ 0.383) that anchors the model's mean total on the WC fixtures to a conservative recent-WC base rate (2.81). 1X2-preserving; backward-compatible (default-off flag).
- Verified impact (73 played matches, 2026-06-11- $\rightarrow$ 06-28): realized mean total 2.959; model OLD 2.441 (bias $\approx$ 0.518) $\rightarrow$ NEW 2.917 (bias $\approx$ 0.042) - the anchor removes $\sim$ 92% of the under-forecast.
- What it changes in outputs: total xG, Over 2.5 and BTTS rise on every fixture; the 1X2 is $\sim$ invariant (it's a totals correction, not a 1X2 change). Edge it unlocks is in goal-derived match-event markets, not liquid winner markets.
- Status: all work is on branch `integrate/fixes-2026-06-30` (full suite 2052 passed, 2 skipped). Nothing is on main/production yet - so the live model still under-forecasts until merged.

2. The xG/totals fix - before vs after (verified)

Remaining-knockout fixtures, neutral venue. OLD = production level (μ 0.205); NEW = WC-anchored (μ 0.383). Computed by running the real fitted model both ways (`contrast.csv`).

Fixture	Total xG (old- $\rightarrow$ new)	P(Over 2.5)	P(BTTS)
France v Sweden	2.42 $\rightarrow$ 2.89	43% $\rightarrow$ 55%	45% $\rightarrow$ 54%
Mexico v Ecuador	2.03 $\rightarrow$ 2.43	33% $\rightarrow$ 44%	41% $\rightarrow$ 50%
England v DR Congo	2.44 $\rightarrow$ 2.92	44% $\rightarrow$ 56%	32% $\rightarrow$ 38%
Spain v Austria	2.64 $\rightarrow$ 3.16	49% $\rightarrow$ 61%	44% $\rightarrow$ 52%
Argentina v Cape Verde	2.62 $\rightarrow$ 3.13	49% $\rightarrow$ 60%	30% $\rightarrow$ 36%
Belgium v Senegal	2.36 $\rightarrow$ 2.82	42% $\rightarrow$ 54%	46% $\rightarrow$ 54%
Portugal v Croatia	2.32 $\rightarrow$ 2.78	41% $\rightarrow$ 52%	46% $\rightarrow$ 55%
USA v Bosnia & Herz.	2.30 $\rightarrow$ 2.75	40% $\rightarrow$ 52%	46% $\rightarrow$ 55%
Mean	2.39 $\rightarrow$ 2.86	42.8% $\rightarrow$ 54.3%	-





1X2 is preserved: the mu shift adds the same constant to both teams' log-rates, so the supremacy log-ratio is exactly invariant ($d = 3.3e-16$); the matrix-implied 1X2 moves $\leq \sim 4pp$ only via the non-linear low-score (tau) correction.

Anchor context (verified historical WC means, martj42_cleaned): WC 2018+2022 = 2.833; since-2010 = 2.864 ($n=3801$); all pre-2026 = 2.907. The 2.81 target is a deliberately conservative choice, slightly below these.

Caveat - is the anchor durable?

It re-applies on every refit (not a frozen number), so it keeps enforcing 2.81. But it's a static stopgap: it targets a *fixed* 2.81, not the live level (realized WC2026 = 2.96), and doesn't learn beyond the negligible drift of refitting (90 WC rows in $\sim 49,500$). The 0.04 was measured on group-stage matches and partly relies on host advantage lifting host games above 2.81 - as hosts are eliminated in the knockouts and the scoring regime shifts, the bias will likely re-open. Durable fix = the F7 opponent-adjusted decay+long-term blend (self-updates with results), or making the anchor target a rolling recent rate. Treat the anchor as a band-aid; validate F7 OOS before sizing off it.

3. Under-the-hood changes this session (all on integrate/fixes-2026-06-30)

Change	What	Status
A1 xG/totals anchor	mu 0.205->0.383; shared apply_wc_level_anchor; default-off flag wc2026_results.json 31->73	green
F3 stale results	matches (authoritative martj42_cleaned); new builder, wired into clean-results odds_snapshots reads now	green
F4 snapshot staleness guard	age-checked (was read 7-day-stale unguarded at accas.py:1302) removed hardcoded	green
F5 honest exposure metrics	p_profit/p_win_50 constants; currency-coherent best/worst (GBP vs USD split)	green
F6 live-ledger exposure	bet_recs.json open exposure from the live ledger: n_open 59->8 opponent-adjusted	green
F7 goal-blend	tournament-decay + long-term DC blend - flag-gated, tracking-only, OOS-gated new ShotsOnTargetModel +	green (+13 tests)
F8 match-event models	FoulsModel; corners de-hardcoded; cards refit - fallback-guarded	green (+24 tests)

Change	What	Status
Durable closing-line capture	DB-less odds_price_history.jsonl mirror + idempotent ingest + fail-loud workflow	green
A6 docs	corrected stale "feed revoked" claims	green

Full suite: 2052 passed, 2 skipped. Backward-compatible: F7/F8 default off ? production bit-identical until validated.

4. Integrity audit - fabricated/stale data found (12 high-risk, all verified)

Issue	Where	Fixed by
xG fix not in production	card.py:604-610 (no anchor)	A1 (on branch; needs merge)
wc2026_results.json stale (31, frozen 06-20) drives settlement + metrics	predledger/settle.py, winrate.py, rigor/build.py	F3
odds_snapshots 7 days stale, read unguarded	accas.py:1302, closecapture.py	F4
hardcoded p_profit/p_win_50 shown as live risk	exposure_dashboard.py:88,91	F5
best/worst mix GBP+USD, labelled GBP	exposure_dashboard.py:75-84	F5
bet_recs.json stale 59-bet exposure vs real 8	site/bet_recs.json	F6
players.json scorer shares all analyst_estimate (238)	data/players.json	data-gated (labelled, not fixed)

The only fabrication that had reached a deliverable was the earlier PDF's "realized 3.000 / n=31" (a stale-file artifact) - corrected here. Adversarial verifier reproduced all 12 findings; none overstated.

5. Model vs market

The model is a good 1X2 calibrator (ECE 0.083, Brier 0.150, n=825) but not a winner-market alpha source: its large disagreements with the liquid Polymarket book - Brazil 18.8% model vs 7.1% market to win; France 12.9% vs 28% - are model error, not edge (the book is efficient: Winner sums to 0.985). The repo's own backtest (n=64) shows no confident edge over market-only. Edge lives in less-efficient match-event markets + execution/cost, which is what A1/F7/F8 target.

6. Conditional bracket (as-if-open)

Real fitted model, 50k sims, conditioned on the 4 played R32 results. Pure-model MAP champion Spain (marginal title leader is Brazil 18.8%, but Brazil loses the modal SF coin-flip to Argentina ~0.50). P(perfect-from-here) ~ 1.7e-5 (~1 in 60,000); risk concentrated in 5 near-coin-flip nodes. A *market-anchored* completion would instead crown France (28% favourite) - consistent with \$5.

7. OddsAPI credit usage

3,793 used / 96,207 remaining of ~100k monthly (verified live). Cost = 1 credit x markets x regions per call; /sports free. Dominant spender: the snapshot daemon (315 capture cycles, 06-11->06-23, markets h2h/totals/btts/h2h_lay) - then the hourly scores cron (h2h, 1 credit/call) and the daily-card scorer-event fetches. Capture stopped 06-23 (daemon died, since restarted), so the last week burned little. *(No per-request credit log exists, so per-source split is the cost-model x observed cadence, not an itemized ledger.)* Runway is ample; finer near-KO capture for true closing lines will raise burn (a budget governor exists in pollsched).

8. Caveats & what is NOT done

- Nothing merged to main/pushed - all on integrate/fixes-2026-06-30 (push + PR pending).
- F7 goal-blend and F8 models are tracking-only / OOS-gated - they do not size real money until they clear an out-of-sample CLV gate (per the project's own rule). Shrinkage knobs are design defaults, not validated.
- Squad adjustment is data-gated (squads.json covers 2/48 teams; players.json is analyst-estimated shares) - implemented as an honest no-op hook, no fabricated squad strengths.
- prop_priors.csv absent in a fresh checkout (gitignored local artifact) - F8 models run on documented fallback constants until the pipeline produces it.
- Anchor durability caveat (\$2): expect the ?0.04 to re-open in the knockouts.

9. Where things live

- This report: docs/research/wca_alpha_2026/MODEL_REVIEW_2026-06-30.{md,pdf}.
- Also: SESSION_REPORT_2026-06-30.{md,pdf}, INTEGRITY_AUDIT_AND_GOALMODEL.md, SWARM_LEDGER.md, the 00-15 dossier.
- Branch integrate/fixes-2026-06-30 (+13 commits ahead of main, full suite green, not pushed).